



Diabetes & Metabolic Syndrome: Clinical Research & Reviews

journal homepage: www.elsevier.com/locate/dsx



Clinical effectiveness of krill oil supplementation on cardiovascular health in humans: An updated systematic review and meta-analysis of randomized controlled trials

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ARTICLE INFO

Keywords:

Blood pressure
CVD
Lipid
Krill oil
Meta-analysis

ABSTRACT

Background: The potential role of krill oil (KO) supplementation on cardiovascular health are inconsistent in several clinical trials. Therefore, our present meta-analysis aimed to systematically evaluate the impacts of supplementation of KO on cardiovascular disease risk factors (CVDRFs).

Methods: Intervention trials assessing KO supplementation on cardiovascular disease (CVD) outcomes were systematically retrieved for pooling. The primary outcome was lipid profile. Secondary outcomes were consisted by blood pressure, glycemic indices, body composition together with inflammatory markers. We synthesized the effect sizes with 95% confidence intervals and weighted mean difference. To explore the heterogeneity source, we employed meta-regression and subgroup analysis. Quality assessment, publication bias, sensitivity-analysis and the certainty of evidence were also carried out.

Results: We included 14 trials (18 treatment arms) with 1458 participants. KO supplementation had beneficial effects on total cholesterol ($P = 0.01$), low-density lipoprotein cholesterol ($P = 0.006$), and triglycerides ($P = 0.0005$). However, no effects were found for other CVDRFs, such as blood pressure, glycemic control, body composition as well as inflammatory markers. Subgroup analyses indicated that these notably favorable effects were observed in trials with a parallel design, treatment duration < 8 weeks and subjects with baseline body mass index $< 28 \text{ kg/m}^2$. The above findings remained consistent in the sensitivity analysis, without obvious publication bias detected.

Conclusions: The current evidence demonstrated that daily KO supplementation may as a candidate for lipid management strategies. In future, studies should pay attention to the relationships of KO intake with the incidence of CVD events or all-cause mortality.

1. Introduction

Cardiovascular disease (CVD) is significantly connected to most disabilities, morbidities, and mortalities in the world [1]. The global

increase in aging populations is attributed to the rapid escalation in the prevalence of CVD, which accounted for more than 17.9 million deaths globally in 2019, and the number of CVD related fatalities would rise to over 23.6 million by 2030 [2]. The significance of public health

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<https://doi.org/10.1016/j.dsx.2023.102909>

Received 28 June 2023; Received in revised form 23 October 2023; Accepted 14 November 2023

Available online 23 November 2023

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necessitates prioritizing primary and secondary prevention for cardiovascular diseases, along with the delay of occurrence and progression of CVD related issues due to its health and financial burden on society. Previous evidences have shown that controlling the CVD risk factors (CVDRFs), such as inappropriate diet, inflammation, type 2 diabetes (T2D), obesity, hypertension, insulin resistance, and dyslipidemia, can decrease or prevent the CVD development [3–5]. Abundant of evidence suggests that the prevention and management of CVD can be controlled through dietary modifications or nutraceutical therapies [6–8]. Long-chain n-3 fatty acids, including eicosapentaenoic acid (EPA) as well as docosahexaenoic acid (DHA), have been proved to lower circulating lipid levels and the risk of cardiovascular events [9,10].

Krill oil (KO), an edible oil extracted from krill (*Euphausia superba*), is natural sources of n-3 polyunsaturated fatty acids. It is abundant in phospholipids, flavonoids, EPA, astaxanthin, DHA, vitamins, trace elements, as well as other substances [11]. In KO, over 80% of EPA and DHA are found in the phosphatidylcholine form [12]. The advantage of KO supplementation encompasses the provision of not only EPA and DHA, but also choline, an essential component for the synthesis of neurotransmitters (acetylcholine) and phospholipids [12,13]. Numerous studies have found that KO supplementation has various health benefits for hyperlipidemia, chronic inflammation, premenstrual syndrome, osteosarcoma, rheumatoid arthritis, metabolic syndrome, nonalcoholic fatty liver disease, and atherosclerosis [13–17]. The previous systematic review and meta-analysis conducted by Ursoniu et al. [18] including only 7 eligible trials with 662 participants showed that KO supplementation can reduce low-density lipoprotein cholesterol and triglycerides in humans. However, the relationship of KO consumption with other cardiovascular risk factors in body composition, glycemic indices, and inflammation were not evaluated in the study. Moreover, owing to the small sample sizes, different supplementation dosages, and diverse underlying health states, the clinical evidence and the recommendation's strength for KO on each CVDRFs has remained unclear.

Here, we were aimed to quantitatively assess the precise effectiveness of daily KO supplement on CVDRFs related to lipid profile, blood pressure, body composition, glycemic indices, and inflammation in humans through a comprehensive review of the current available literature as well as a meta-analysis of available randomized controlled trials (RCTs).

2. Methods

2.1. Strategy for literature search

We conducted the design, implementation, and reporting complying with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Firstly, two authors (H.H. and D.L.) carried out the keyword-based search in Cochrane Central Register of Controlled Trials (CENTRAL), PubMed, as well as EMBASE through March 2023, to find all eligible RCTs in human subjects regarding the clinical effectiveness of krill oil supplementation on CVDRFs. The following Medical Subjects Headings (MeSH) and free-text terms were used: (“krill oil” OR “euphausiaceae” OR “euphausia superba”) AND (“cardiovascular risk” OR “cardiovascular disease” OR “CVD” OR “cardiovascular event” OR “atherosclerosis” OR “coronary artery disease”

OR “acute coronary event” OR “ischemic heart disease” OR “endothelial” OR “overweight” OR “obesity” OR “waist circumference” OR “body mass index” OR “coronary heart disease” OR “myocardial infarct” OR “blood pressure” OR “arterial pressure” OR “C-reactive protein” OR “inflammatory” OR “lipid” OR “triglycerides” OR “cholesterol” OR “hypertension” OR “insulin resistance” OR “insulin sensitivity”). Related reviews and references cited by the included papers were also screened to find more relevant papers.

2.2. Study selection criteria and data extraction

The Patient, Intervention, Comparators, Outcome, and Study Design (PICOS) approach were employed to select eligible studies for pooling (Table 1). Data were retrieved by two researchers (G.Z. and B.H.). Non-agreements were resolved by a third researcher (D.L.). The following characteristics of included studies were extracted:

- **Publication data:** first author, year of publication, and country in which the trial was carried out.
- **Participants’ characteristics:** mean age, health condition, and gender.
- **Study characteristics:** Enrolled and completed patient number, KO dosage and formulation, control treatment type, intervention duration, details of the study design.
- **Study outcomes:** The primary outcomes were triglycerides (TG), high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C), as well as total cholesterol (TC). Secondary outcomes included C-reactive protein (CRP), waist circumference (WC), body weight (BW), hip circumference (HC), body mass index (BMI), insulin, homeostatic model assessment of insulin resistance (HOMA-IR), fasting blood glucose (FBG), HbA1c, diastolic blood pressure (DBP), as well as systolic blood pressure (SBP). Baseline and post-treatment means together with standard deviations (SDs) of primary outcomes or their difference between baseline and post-treatment were captured in these enrolled studies. Based on the Cochrane Handbook for Systematic Reviews of Interventions, SDs were estimated using the standard error (SE) or 95% confidence intervals (CIs) when they were not directly available [19]. The same variable published in various units across studies would be translated to data in identical units. For trials with several intervention groups (i.e., different dosages of intervention), all the experimental groups would be grouped together and compared with the control group, respectively. Lastly, endpoint data from the same cohort would be collected from the report with the greatest follow-up duration or the most substantial sample size.

2.3. Quality assessment

We applied the Cochrane risk-of-bias tool for RCTs. Criteria, such as incomplete outcome data, blinding of outcome assessment, random sequence generation, selective reporting, blinding of participants and personnel, as well as allocation concealment, were applied to evaluate the characteristics’ quality. According to the suggestions from the Cochrane Handbook, a qualitative degree (no, yes, or unclear) was assigned for each item, and judgment of each included study was labeled

Table 1
PICOS criteria for inclusion of studies.

Parameter	Description
Population (P)	Adults participants (age >18 years) with healthy or otherwise
Intervention (I)	Subjects needed to have specifically ingested krill oil supplementation. Intervention duration of at least 2 weeks
Comparator (C)	Placebo interventions or other oil products interventions
Outcomes (O)	The study provided sufficient data on at least one of the cardiovascular health outcomes, included cardiovascular disease risk factors related to lipid profile, blood pressure, body composition, glycemic indices, and inflammation
Study design (S)	Original randomized controlled trials with a parallel or crossover design

with “unclear”, “high”, or “low” risk of bias [19].

2.4. Statistical analysis

DerSimonian-Laird random-effects model were used to calculate the 95% CIs as well as the weighted mean difference (WMD) for continuous outcomes. The I^2 index and the Q test evaluated the heterogeneity between trials. I^2 scores $\geq 50\%$ indicated moderate to high heterogeneity [20]. A conventional analysis of sensitivity was conducted by removing one study each time to examine whether the overall results relied on individual studies. Then, we carried out analysis of pre-specified subgroup to explore the overall effects of KO in CVDRFs and the probable causes of heterogeneity, including study design (crossover vs. parallel design), baseline BMI (<28 vs. ≥ 28 kg/m²), KO dose (<3 vs. ≥ 3 g/d), and the intervention duration (<8 vs. ≥ 8 weeks). The publication bias probability was examined using the asymmetry of Egger's weighted regression statistics, Begg's rank correlation, as well as funnel plot [21, 22]. We used a random-effects meta-regression analysis on the basis of the restricted maximum likelihood (REML) to determine if the duration and amount of KO supplementation were related to the heterogeneity source [23]. Subgroup analyses, meta-regression, as well as publication bias assessment were carried out only for lipid profile considering small numbers of studies for the secondary outcomes ($n \leq 5$). Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach was adopted for the evaluation of the quality of evidences for every metabolic and cardiovascular biomarker. The evidence quality was classified into very low, low, moderate, and high [24]. RevMan (v. 12.0) and STATA (v. 5.0) were applied for all statistical analyses. Two-sided comparisons were applied in all analyses unless stated otherwise. A significant difference was set as a $P < 0.05$.

3. Results

3.1. Trials identification and selection

Based on the search strategy, 163 records were initially searched out. Four additional ones were found by screening the reference lists. Eighty records were excluded for duplication analysis, and after reviewing the titles or abstracts or full text, 19 publications were excluded for various reasons, such as reviews, or meta-analyses and systematic review, studies that did not provide adequate data for the outcomes, did not report the primary outcomes, bioavailability study, or used mixture of KO and other nutrients. The work conducted by Berge et al. [25] were grouped into 4 treatment arms (0.5 g/d, 1 g/d, 2 g/d, 4 g/d), and the work conducted by Ramprasath et al. [26] were grouped into 2 treatment arms (1.5 g/d, 3 g/d). Finally, 14 publications (18 treatment arms) composed of 1458 participants were encompassed in the current meta-analysis to evaluate KO's effects on cardiovascular health [25–38]. Flowchart of the selection strategy are shown in Fig. 1.

3.2. Subjects and study characteristics

The publication date of the included trials ranged from 2009 to 2022. Researchers from eight countries performed these studies: the United Kingdom ($n = 1$) [27], Norway ($n = 3$) [25,33,38], Brazil ($n = 1$) [28], USA ($n = 2$) [29,30], Japan ($n = 1$) [37], Canada ($n = 2$) [26,32], Finland ($n = 1$) [34], as well as Australia ($n = 2$) [35,36]. The study conducted by Mozaffarian et al. [31] enrolling the participants at US, Canadian, and Mexican centers. In each trial, the number of participants ranged from 10 to 520, and 1458 individuals were finally included for data pooling. Two studies included only female subjects [28,36], twelve included both genders [25–27,29–35,37,38]. The average age ranged from 28.23 to 71.1, and the BMI values ranged between 23.67 and 32.95 kg/m². Healthy subjects were enrolled in six studies [26,27,32,33,36, 38], and overweight subjects were enrolled in 2 studies [28,30]. Other trials were carried out among participants with various diseases,

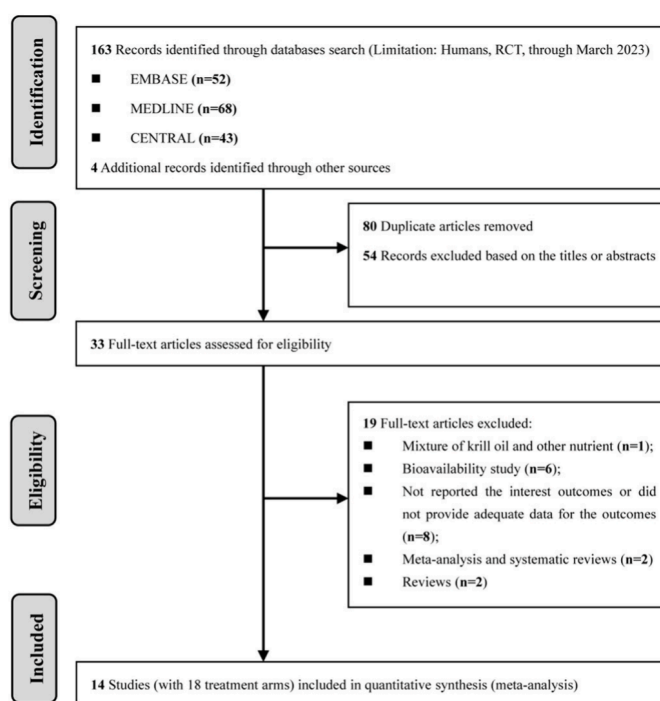


Fig. 1. Flow diagram for identification of relevant studies.

Pubmed, <http://www.ncbi.nlm.nih.gov/pubmed/>; Embase, <http://www.embase.com/home>; Cochrane Library, <https://www.cochranelibrary.com/#> The work conducted by Berge et al. were separated into 4 treatment arms (0.5 g/d, 1 g/d, 2 g/d, 4 g/d), and the work conducted by Ramprasath et al. were separated into 2 treatment arms (1.5 g/d, 3 g/d).

including T2D ($n = 1$) [29], hypertension ($n = 1$) [34], hypercholesterolemic ($n = 2$) [25,31], and knee osteoarthritis or knee pain ($n = 2$) [35,37]. The doses of KO administered ranged from 0.5 to 5 g/day. The control interventions included mixed vegetable oil, olive oil, mineral oil capsule, cornstarch, corn oil, high-oleic sunflower oil, maize starch, or safflower oil. The intervention duration varied between 4 and 26 weeks. Seven studies used a parallel-group design [25,27,28,31,33,37,38], and others used a crossover design [26,29,30,32,34–36]. Thirteen trials explored the role of KO in lipid profile [25–27,29–38], three in BP [26, 27,33], five in glycemic indices [27,29,30,33,36], three in body composition [26–28], and two in inflammatory markers [27,30]. Study characteristics and the main participants' information are presented in Table 2.

Fig. 2 summarized the Cochrane Collaboration tool's risk of bias assessment, with an unknown risk in 10 studies [25,26,28,30–33, 36–38], and a low risk in 4 studies [27,29,34,35]. Thus, of the studies analyzed, 4 were deemed to be of “high quality”, whereas the rest were categorized as “fair quality”.

3.3. Primary outcome: effects of KO supplementation on lipid profile

Pooled results of a random-effects model suggested that KO supplementation decreased the levels of TC (WMD, - 4.18 mg/dL; 95% CI, - 7.44 to - 0.93; $P = 0.01$; $I^2 = 0\%$), LDL-C (WMD, - 3.76 mg/dL; 95% CI, - 6.47 to - 1.06; $P = 0.006$; $I^2 = 0\%$), as well as TG (WMD, - 10.76 mg/dL; 95% CI, - 16.84 to - 4.69; $P = 0.0005$; $I^2 = 85\%$) with statistical significance. However, the overall analysis showed that KO did not affect HDL-C (WMD, 0.12 mg/dL; 95% CI: - 1.09 to 1.34; $P = 0.84$; $I^2 = 0\%$). A forest plot for the impacts of KO on lipid profile is shown in Fig. 3.

Subgroup analyses were also performed to provide further insight into the impact of KO on lipid profile parameter in various pre-defined subgroups. Table 3 summarized the results of subgroup analysis. In terms of intervention durations, significant effects on TC (WMD, - 4.35

Table 2

Characteristics of study populations, type of interventions, and study designs in the included trials.

Study	Study design	Location	Sample size (enrolment/completed)	Sex (M/F)	Study participants	Mean age ²	BMI (kg/m ²)	Intervention form		Dose (g/d)	Duration (week)	Outcomes
								Treatment Group	Control Group			
Alkhedhairi et al. [27]	R, DB, P	United Kingdom	102/94	41/53	Healthy older adults	71.1	25.85	Krill oil capsules	Mixed vegetable oil	4	24	TC, LDL-C, HDL-C, TG, FBG, Insulin, CRP, BW, SBP, DBP
Berge et al. (2014) [25]	R, DB, P, PC	Norway	300/267	184/83	Adults with borderline high or high triglyceride levels	44.04	29.6	Krill oil capsules	Olive oil	0.5, 1, 2, 4	12	TC, LDL-C, HDL-C, TG
Lobo et al. [28]	DB, P, PC	Brazil	38/26	0/26	Overweight Women	NA	31.05	Krill oil capsules	Mineral oil capsule	3	8	BMI, WC
Lobraico et al. [29]	R, DB, C, PC	USA	48/47	31/16	Participants with type 2 diabetes	64.8	30.5	Krill oil capsules	Olive oil	1	4	TC, LDL-C, HDL-C, TG, FBG, HbA1c, HOMA-IR
Maki et al. [30]	R, DB, C	USA	50/50	8/42	Overweight and obese men and women	48.4	32.95	Krill oil capsules	Olive oil	2	4	TC, LDL-C, HDL-C, TG, FBG, Insulin, HOMA-IR, CRP
Mozaffarian et al. [31]	R, DB, P, PC	US, Canada, Mexico	520/520	339/181	Patients With Severe Hypertriglyceridemia	54.9	31.5	Novel ω -3 Krill Oil Agent	Cornstarch	4	26	LDL-C, HDL-C, TG
Ramprasath et al. [32]	R, DB, C, PC	Canada	24/24	12/12	Healthy individuals	28.23	23.76	Krill oil capsules	Corn oil	0.6	4	TC, LDL-C, HDL-C, TG
Ramprasath et al. [26]	R, DB, C, PC	Canada	24/20	11/9	Healthy participants	NA	23.67	LPH krill oil capsules HPL krill oil capsules	Corn oil	1.5, 3	4	TC, LDL-C, HDL-C, TG, SBP, DBP, BW, BMI, WC, HC
Rundblad et al. [33]	R, P	Norway	26/24	12/12	Healthy subjects	NA	27.2	Krill oil capsules	High-oleic sunflower oil	4	8	TC, LDL-C, HDL-C, TG, SBP, DBP, FBG, HbA1c, BMI
Sarkkinen et al. [34]	R, DB, C, PC	Finland	35/35	18/17	Subjects with mildly or moderately elevated blood pressure	55.4	28	Krill oil capsules	Maize starch	4	8	LDL-C, HDL-C, TG
Stonehouse et al. [35]	R, DB, C, PC	Australia	235/221	106/129	Adults with mild to moderate knee osteoarthritis	55.9	28.3	Krill oil capsules	Mixed vegetable oil	1	24	TC, LDL-C, HDL-C, TG
Sung et al. [36]	R, C	Australia	17/10	0/10	Healthy women	28.5	25.8	Krill oil capsules	Olive oil	5	4	TC, LDL-C, HDL-C, TG, FBG
Suzuki et al. [37]	R, DB, P, PC	Japan	50/47	6/41	Adults with mild kneepain	64.7	23.85	Krill oil capsules	Safflower oil	2	4	TC, LDL-C, HDL-C, TG
Ulven et al. [38]	R, P	Norway	86/73	27/59	Healthy volunteers	40.4	23.75	Krill oil capsules	No supplementation	3	7	TC, LDL-C, HDL-C, TG

NA, not available; M, male; F, female; R, randomized; PC, placebo-controlled; DB, double-blind; P, parallel; C, crossover; TC, total cholesterol; LDL-C, low-density lipoprotein cholesterol; HDL-C, high-density lipoprotein cholesterol; TG, triglycerides; SBP, systolic blood pressure; DBP, diastolic blood pressure; FBG, fasting blood glucose; HOMA-IR, homeostatic model assessment of insulin resistance; BW, body weight; BMI, Body Mass Index; WC, waist circumference; HC, hip circumference; CRP, C-reactive protein.

mg/dL; 95% CI, - 7.78 to - 0.92; $P = 0.01$), LDL-C (WMD, - 3.68 mg/dL; 95% CI, - 6.97 to - 0.39; $P = 0.03$) and HDL-C (WMD, 2.27 mg/dL; 95% CI, 0.16 to 4.38; $P = 0.04$) were found in subjects with the consumption of KO for <8 weeks; In contrast, a reduction in TG (WMD, - 13.99 mg/dL; 95% CI, - 21.7 to - 6.28; $P = 0.0004$) was found in those with the consumption of KO for ≥ 8 weeks with statistical significance. In terms of dose, supplementation of KO decreased TC (WMD, - 3.95 mg/dL; 95%

CI, - 7.91 to - 0.06; $P = 0.04$) and TG (WMD, - 12.06 mg/dL; 95% CI, - 20.75 to - 3.38; $P = 0.007$) in trials with daily KO dose <3 g/d with statistical significance; a reduction in LDL (WMD, - 4.30 mg/dL; 95% CI, - 7.97 to - 0.62; $P = 0.02$) and TG (WMD, - 8.18 mg/dL; 95% CI, - 12.67 to - 3.69; $P = 0.0004$) were also observed in those with daily KO dose ≥ 3 g with statistical significance. In terms of baseline BMI, the supplementation of KO decreased TC (WMD, - 6.71 mg/dL; 95% CI:

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Alkhedhairi (2022) [27]	+	+	+	+	+	+	+
Berge (2014) [25]	?	?	+	+	+	+	+
Lobo (2022) [28]	?	?	+	+	+	+	+
Lobraico (2015) [29]	+	+	+	+	+	+	+
Maki (2009) [30]	?	?	+	+	+	+	+
Mozaffarian (2022) [31]	?	?	+	+	+	+	+
Ramprasath (2013) [32]	?	?	+	+	+	+	+
Ramprasath (2015) [26]	?	?	+	+	+	+	+
Rundblad (2018) [33]	?	?	+	+	+	+	+
Sarkkinen (2018) [34]	+	+	+	+	+	+	+
Stonehouse (2022) [35]	+	+	+	+	+	+	+
Sung (2018) [36]	?	?	+	+	+	+	+
Suzuki (2016) [37]	?	?	+	+	+	+	+
Ulven (2011) [38]	?	?	+	+	+	+	+

Fig. 2. Quality assessment of the included studies. Labeling an item as “Question mark” indicated an unclear or unknown risk of bias; Labeling an item as “negative sign” indicated an high risk of bias; Labeling an item as “positive sign” indicated an low risk of bias.

- 11.49 to - 1.93; $P = 0.006$), LDL-C (WMD, - 5.43 mg/dL; 95% CI, - 9.65 to - 1.22; $P = 0.01$) and TG (WMD, - 7.47 mg/dL; 95% CI, - 14.64 to - 0.30; $P = 0.04$) in subjects with baseline BMI <28 kg/m² with statistical significance; a reduction in TG (WMD, - 12.15 mg/dL; 95% CI, - 19.82 to - 4.48; $P = 0.002$) was also observed in those with baseline BMI ≥28 kg/m² with statistical significance. In terms of study design, supplementation with KO in trials with crossover design showed

a considerable reduction in TC (WMD, - 4.73 mg/dL; 95% CI, - 8.53 to - 0.94; $P = 0.01$) and LDL-C (WMD, - 4.88 mg/dL; 95% CI, - 8.77 to - 0.98; $P = 0.01$); a reduction in TG (WMD, - 15.68 mg/dL; 95% CI, - 23.29 to - 8.08; $P = 0.00001$) was observed in trials using a parallel design with statistical significance.

3.4. Secondary outcomes: effects of KO supplementation on blood pressure, glycemic indices, body composition as well as inflammatory markers

Table 4 outlines data of secondary outcomes. For the markers of blood pressure, the pooled results from the random-effects model proved that KO intervention did not affect SBP (WMD, 1.07 mmHg; 95% CI, - 1.40 to 3.54; $P = 0.40$; $I^2 = 10\%$) as well as DBP (WMD, 1.48 mmHg; 95% CI, - 0.24 to 3.20; $P = 0.09$; $I^2 = 10\%$). For the markers of glycemic indices, KO did not appear to significantly alter the levels of FBG (WMD, - 0.08 mg/dL; 95% CI, - 0.33 to 0.17; $P = 0.53$; $I^2 = 64\%$), insulin (WMD, 0.24 μU/mL; 95% CI, - 2.13 to 2.60; $P = 0.85$; $I^2 = 0\%$), HbA1c (WMD, - 0.01; 95% CI, - 0.20 to 0.18; $P = 0.93$; $I^2 = 0\%$) as well as HOMA-IR (WMD, - 0.07; 95% CI, - 0.96 to 0.83; $P = 0.88$; $I^2 = 40\%$). For the markers of body composition, KO intake did not affect the BW (WMD, - 0.14 kg; 95% CI, - 0.60 to 0.32; $P = 0.55$; $I^2 = 0\%$), BMI (WMD, - 0.09 kg/m²; 95% CI, - 0.27 to 0.10; $P = 0.35$; $I^2 = 0\%$), WC (WMD, - 0.09 cm; 95% CI, - 0.57 to 0.38; $P = 0.69$; $I^2 = 0\%$), as well as HC (WMD, 0.29 cm; 95% CI, - 0.16 to 0.75; $P = 0.20$; $I^2 = 0\%$). We also evaluated the effects of KO ingestion on inflammation markers, such as CRP. The present result shown that KO supplementation was not related to a significant decrease in CRP (WMD, -0.21 mg/L; 95% CI, - 1.38 to 0.96; $P = 0.73$; $I^2 = 0\%$).

3.5. Clinical safety evaluation

In the included trials, only a few cases with mild to moderate gastrointestinal symptoms, such as flatulence, heartburn and diarrhea were found. The absence of serious adverse events following KO consumption suggested a high level of tolerance and safely.

3.6. Sensitivity analyses

The result's robustness was verified using the leave-one-out method based on sensitivity analyses. The pooled results or overall effect size for lipid profiles remained solid and were not sensitive to any individual study (Supporting materials Fig. S1). Additionally, after employing the fixed-effects model, the analysis of sensitivity revealed that the combined effects of KO on lipid profiles did not significantly change.

3.7. Meta-regression between doses and duration of KO intake and lipid profile

The correlation of changes in lipid profile with KO dosage (g/day) and duration (weeks) were evaluated in this study. No obvious differences were observed between the KO duration and the influence on TG, HDL-C, LDL-C, as well as TC (Fig. 4). Meanwhile, the KO amount consumed per day was not related to changes in TG, HDL-C, TC, as well as LDL-C (Fig. 5).

3.8. GRADE evidence profile assessment

The GRADE profiles for the clinical evidence of KO supplementation on blood lipids, BP, glycemic indices, inflammatory markers, and BW management are presented in Supporting materials Table S1. In the GRADE Working Group evidence level rates, we found low quality for HOMA-IR; moderate quality for TG, SBP, DBP, FBG, Insulin, HbA1c, BW, BMI, WC, HC, and CRP; and high quality for TC, LDL-C, and HDL-C.

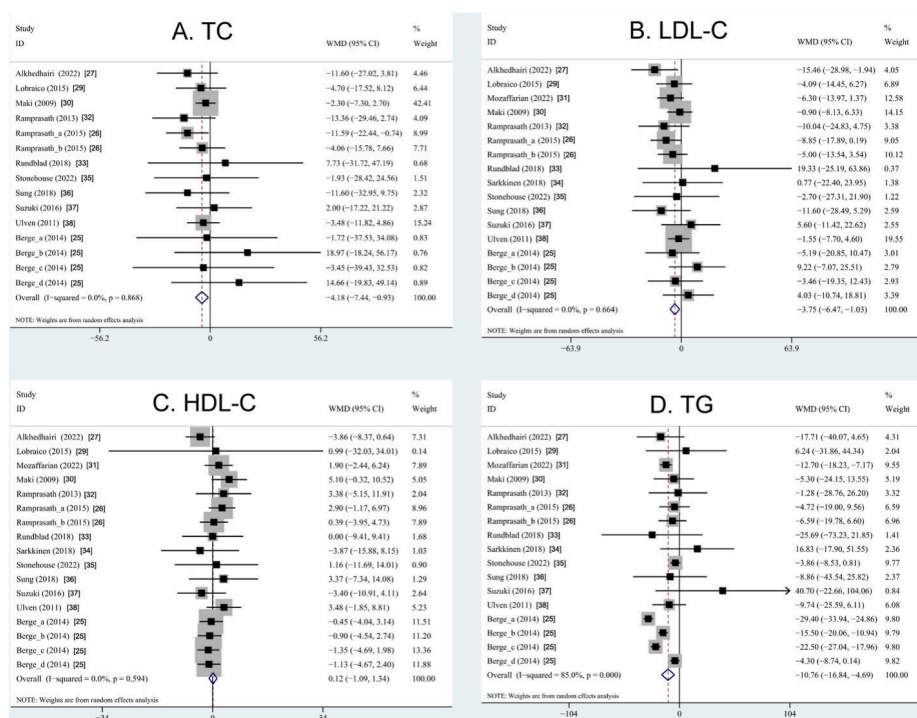


Fig. 3. Forest plot illustrating the weighted mean difference of krill oil on blood lipids with a random effects model. TC, total cholesterol; LDL-C, low-density lipoprotein cholesterol; HDL-C, high-density lipoprotein cholesterol; TG, triglycerides.

Table 3

Subgroup analyses of TC, LDL-C, HDL-C and TG stratified by previously defined study characteristics.

Variables	Arms	TC (mg/dL)				Arms	LDL-C (mg/dL)				Arms	HDL-C (mg/dL)				Arms	TG (mg/dL)			
		WMD (95% CI) ¹	P	I ² (%)			WMD (95% CI) ¹	P	I ² (%)			WMD (95% CI) ¹	P	I ² (%)			WMD (95% CI) ¹	P	I ² (%)	
Duration																				
≥8 weeks	7	- 2.63 (- 12.94 to 7.68)	0.62	0	9	- 3.92 (- 8.72 to 0.89)	0.11	1	9	- 0.94 (- 2.43 to 0.55)	0.21	0	9	- 13.99 (- 21.70 to - 6.28)	0.0004*	92				
< 8 weeks	8	- 4.35 (- 7.78 to - 0.92)	0.01*	0	8	- 3.68 (- 6.97 to - 0.39)*	0.03	0	8	2.27 (0.16–4.38)	0.04*	0	8	- 5.32 (- 12.30 to 1.66)	0.13	0				
Dose																				
≥3 g/day	6	- 4.63 (- 10.45 to 1.19)	0.12	0	8	- 4.30 (- 7.97 to - 0.62)	0.02*	0	8	- 0.11 (- 1.95 to 1.72)	0.90	0	8	- 8.18 (- 12.67 to - 3.69)	0.0004*	20				
< 3 g/day	9	- 3.98 (- 7.91 to - 0.06)	0.04*	0	9	- 3.13 (- 7.14 to 0.88)	0.13	0	9	0.31 (- 1.32 to 1.94)	0.71	0	9	- 12.06 (- 20.75 to - 3.38)	0.007*	89				
Baseline BMI																				
≥28 kg/m ²	7	- 2.00 (- 6.44 to 2.44)	0.38	0	9	- 2.31 (- 6.19 to 1.57)	0.24	0	9	- 0.14 (- 1.68 to 1.40)	0.86	0	9	- 12.15 (- 19.82 to - 4.48)	0.002*	92				
< 28 kg/m ²	8	- 6.71 (- 11.49 to - 1.93)	0.006*	0	8	- 5.43 (- 9.65 to - 1.22)	0.01*	11	8	0.57 (- 1.60 to 2.75)	0.61	11	8	- 7.47 (- 14.64 to - 0.30)	0.04*	0				
Study Design																				
Crossover	7	- 4.73 (- 8.53 to - 0.94)	0.01*	0	8	- 4.88 (- 8.77 to - 0.98)	0.01*	0	8	2.32 (- 0.01 to 4.65)	0.05	0	8	- 3.87 (- 7.85 to 0.12)	0.06*	0				
Parallel	8	- 2.66 (- 8.98 to 3.66)	0.41	0	9	- 2.58 (- 6.95 to 1.78)	0.25	14	9	- 0.70 (- 2.13 to 0.73)	0.34	0	9	- 15.68 (- 23.29 to - 8.08)	0.00001*	89				

TG, triglycerides; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; TC, total cholesterol; WMD, weighted mean differences; CI, confidence interval; BMI, body mass index; *Indicates a significant result.

Table 4
Effect of KO supplementation on secondary outcomes.

Outcomes	No. of treatment arms	No. of patients	WMD (95% CI)	P Value	I ² , %	P Value of Heterogeneity
SBP	4	206	1.07 (- 1.40 to 3.54)	0.40	10	0.34
DBP	4	206	1.48 (- 0.24 to 3.20)	0.09	0	0.76
BW	4	208	- 0.14 (- 0.60 to 0.32)	0.55	0	0.54
BMI	4	130	- 0.09 (- 0.27 to 0.10)	0.35	0	0.76
WC	3	103	- 0.09 (- 0.57 to 0.38)	0.69	0	0.40
HC	2	80	0.29 (- 0.16 to 0.75)	0.20	0	0.70
FBG	5	290	- 0.08 (- 0.33 to 0.17)	0.53	64	0.02
Insulin	2	152	0.24 (- 2.13 to 2.60)	0.85	0	0.46
HbA1c	2	118	- 0.01 (- 0.20 to 0.18)	0.93	0	0.77
HOMA-IR	2	144	- 0.07 (- 0.96 to 0.83)	0.88	40	0.20
CRP	3	176	- 0.21 (- 1.38 to 0.96)	0.73	0	0.58

TG, triglycerides; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; TC, total cholesterol; SBP, systolic blood pressure; DBP, diastolic blood pressure; FBG, fasting blood glucose; HOMA-IR, homeostatic model assessment of insulin resistance; BW, body weight; BMI, Body Mass Index; WC, waist circumference; HC, hip circumference; CRP, C-reactive protein; WMD, weighted mean differences; CI, confidence interval.

3.9. Publication bias assessment

Based on the funnel plots, we did not discover any significant publication bias in the lipid profile outcomes (Fig. 6). Furthermore, consistent results were obtained using the tests of Begg rank correlation and Egger linear regression ($P > 0.05$) (Supporting materials Table S2).

4. Discussion

According to the literature from RCTs and epidemiology, natural dietary products and nutraceutical therapies significantly affect CVD prevention and atherosclerosis-mediated cardiovascular events for many reasons, such as poor adherence to medications, drug contraindications, various side effects, or individual preference for natural or alternative therapies [39,40]. In the present study, we summarized the available scientific data on the beneficial effects of KO ingestion on the biomarkers associated with CVD risk acquired in RCTs. For statistical pooling, we included 14 trials (18 treatment arms) with 1458 individuals, and found that KO positively affects TC, LDL-C as well as TG. Despite of that, compared to control group, the HDL-C, SBP, DBP, FBG, Insulin, HOMA-IR, BW, BMI, WC, HC, and CRP are not changed after KO supplementation treatment. KO demonstrated a favorable tolerability profile, showing no substantial difference in adverse events between KO and control treatments. According to our subgroup analysis, KO intake dramatically suppressed TC, LDL-C as well as TG levels in the following subgroups: KO < 3 g/d, mean BMI < 28 kg/m², duration < 8 weeks, and

crossover design. In addition, the subgroup analysis findings showed that KO consumption lowered TG levels in trials with duration ≥ 8 weeks, dose ≥ 3 g/d, and parallel design with statistical significance. Our findings may provide novel insight that could be integrated into dietary strategies aimed at preventing atherosclerosis development, promoting cardiovascular health in human beings.

Hypercholesterolemia is a prognostic factor for atherosclerotic heart disease. The escalated LDL-cholesterol levels may cause plaque formation through macrophage foam cells induced by LDL-cholesterol oxidation, resulting in atherosclerosis [41]. As a result, the maintenance of proper cellular function together with the prevention of atherosclerotic cardiovascular disease is dependent on cholesterol homeostasis [42]. Epidemiological and prospective studies have robustly established the favorable impact of LDL-C reduction, as evidenced by a proportional decrease in cardiovascular disease events corresponding to each 1% decrease in LDL-C levels [43]. In accordance with the updated guidelines from the National Cholesterol Education Program (NCEP) regarding cholesterol testing and management, the primary objective of drug therapy prioritizes progressively lower LDL-C levels for the treatment of cholesterol and prevention of cardiovascular disease [43]. As evidenced by our meta-analysis, KO might be used as an adjunct to pharmacological therapy in patients with dyslipidemia. Although the exact mechanisms responsible for the lipid-lowering impacts of KO remain obscure. Several mechanisms underlying the antihyperlipidemic impact of KO were as follows: First, KO can reduce the synthesis of cholesterol by reducing the activity of HMG-CoA reductase through

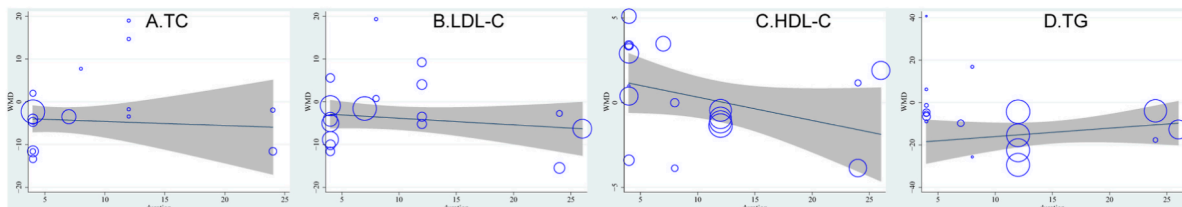


Fig. 4. Meta-regression plots of the association between mean changes in lipid profiles and duration of supplementation with krill oil. The size of each circle is inversely proportional to the variance of change.

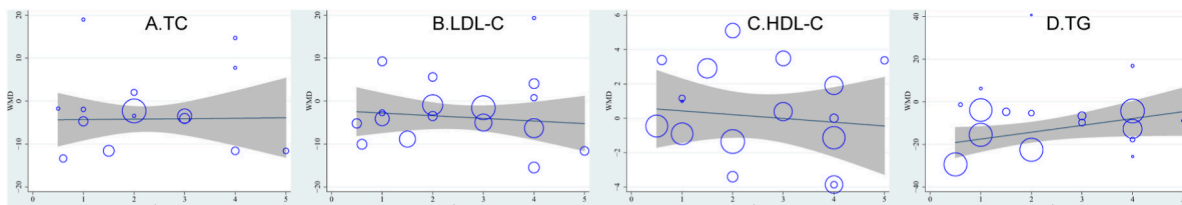


Fig. 5. Meta-regression plots of the association between mean changes in lipid profiles and dose of supplementation with krill oil. The size of each circle is inversely proportional to the variance of change.

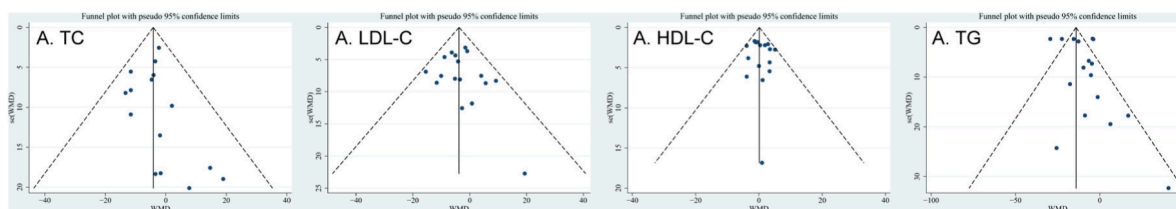


Fig. 6. Funnel plot demonstrating publication bias in the studies reporting the impact of krill oil on lipid profiles.

phosphorylation of AMPK [44,45]. Second, KO supplementation can lower serum TC or LDL-C level in cases with hypercholesterolemia via increasing LDL-C absorption into tissue and through cholesterol esterification [46]. Finally, KO treatment has a potential in increasing the fecal output of cholesterol and bile acid, resulting in increased cholesterol excretion based on the blockade of reabsorption and promotion of bile excretion [47].

Some investigators also found that dietary or supplementation with KO prevents weight gain in an obese animal model and improves the metabolic syndromes from insulin resistance by inhibiting the hepatic steatosis and the gene expressions involving glucose production [48–50]. KO reduced blood glucose levels by boosting glucokinase activity for glucose consumption and decreasing glucose-6-phosphatase and phosphoenolpyruvate carboxykinase activity for gluconeogenesis [51]. The effects of KO on the change of BP, glycemic indices, and body composition have been inconsistent and have remained not fully analyzed. Thus, the role of KO consumption in BP, glycemic indices, and body composition related outcomes were examined, turning out with no significant results.

Two previous systematic review and meta-analysis reported the role of KO supplements in the concentrations of plasma lipid [18,52]. The previous study conducted by Ursoniu et al. [18] including 7 eligible trials with 662 participants showed that KO supplementation can reduce LDL-C and TG in humans. Another network meta-analysis conducted by Kim et al. [52] revealed that KO was associated with significantly lower TG levels than control supplements. However, the net differences in LDL-C, TG, HDL-C, as well as TG were not obviously different in the groups of krill oil and fish oil. Compared to the previous meta-analyses, our current meta-analysis was the most recent, critical, in-depth, comparative investigation, which typically redefined and strengthened earlier study conclusions. First, we included 7 recent published RCT (14 compared with 7) with 796 additional participants (662 compared with 1458) for the present analysis, yielding greater statistical power to our study. Second, this is the first meta-analysis to evaluate evidence from RCTs on the role of KO supplementation in other cardiovascular risk factors related to body composition, glycemic indices, and inflammation. Moreover, differences in the interventions used and inconsistency in the effect and effect size observed across studies contribute to the indirectness of evidence, which can be partly attributed to lenient inclusion criteria. To better assess the robustness of the available evidence and to make more objective conclusions from the results, we performed several subgroup analyses to investigate the effects of KO intervention in terms of study design, baseline BMI, intervention duration, and KO dose. Third, we performed the analyses of sensitivity and publication bias to obtain more objective decisions and did not find substantial publication bias. The dose or time response of KO consumption was examined through linear meta-regression. Finally, we used REML-based dose-response analysis to find an optimal duration and dosage for KO intervention. Additionally, we conducted the criteria of GRADE to evaluate the clinical evidence's overall body and the recommendation's strength for each outcome, different from previous studies.

5. Limitation

Although the present meta-analysis provides useful implications for clinical practice, some certain limitations that should be mentioned. First, a majority of the included trials carried out in relatively heterogeneous populations, including healthy participants, subjects with hypertension, hypercholesterolemic, and knee osteoarthritis or knee pain. Thus, since we did not have access to the original data from the studies reviewed, it was not possible to fully establish the impact of KO on cardiovascular disease risk factors in different populations through subgroup analyses. Second, only a small number of trials were available for some secondary outcomes such as HC, WC, Insulin, HbA1c, HOMA-IR and CRP. Therefore, these findings must be interpreted cautiously, and further more large-scale, high-quality clinical studies are required to reach more confident conclusions. Third, the recruited subjects could have various background habits of consumptions, medications, and lifestyles which may affect the lipid profile, blood pressure, and other CVD risk factors significantly in individuals. Finally, we found significant heterogeneity among trials for the TG outcome ($I^2 = 85\%$). However, sensitivity, meta-regression, and a priori subgroup analyses did not completely solve the sources of heterogeneity.

6. Conclusions

KO supplementation could significantly improve lipid metabolism by decreasing the TC, LDL-C and TG levels. However, KO supplementation show no beneficial effect in SBP, DBP, FBG, Insulin, HbA1c, HOMA-IR, BW, BMI, WC, HC, or CRP. The effects of KO on lipid markers were affected by the trial design, supplement dosage, baseline BMI, and intervention duration. Our subgroup analyses showed that supplementation of krill oil decreased TC and TG in trials with daily KO dose <3 g/d. Moreover, a reduction in LDL-C and TG were also observed in those with daily KO dose ≥ 3 g. Our findings might help healthcare practitioners and the general public better comprehend the evidence on the CVD preventive benefits of KO supplementation. It should be noted that although certain beneficial lipid-modifying effects were observed after KO supplementation, in future, studies should pay attention to relationships of KO intake with the incidence of CVD events or all-cause mortality. Additionally, researches on a standardize KO supplementation protocol, including the dosing regimen and duration of administration, are also of necessity.

Statement of significance

Evidence regarding the potential role of krill oil (KO) supplementation on cardiovascular health in humans are inconsistent. The present updated meta-analysis was the most recent, critical, in-depth comparative investigation, which systematically analyze published clinical evidence and evaluate of the effects of KO supplementation in preventing or treating cardiovascular disease (CVD) in humans. The pooled result showed that KO supplementation could significantly improve lipid metabolism by decreasing the total cholesterol, low-density lipoprotein cholesterol, and triglycerides levels. The present findings might help healthcare practitioners and the general public better comprehend the

evidence on the CVD preventive benefits of KO supplementation.

Credit authorship contribution statement

H. H.: Conception and design of the study, acquisition of data, analysis and interpretation of data, drafted the manuscript, discussed the idea of the meta-analysis, submitted the paper.; D.L. and B. H.: Completed the database searches and selected, reviewed the articles and extracted the data; G.Z. and Y.C.: Reviewed and extracted the data, and performed the data analyses, All authors read and approved the final manuscript.

Funding

This work was supported by the Traditional Medicine Research Program of Guangdong Province under Grant (No. 20211410); Medical Scientific Research Foundation of Guangdong Province under Grant (No. A2021356).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.dsx.2023.102909>.

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